Charged particles in EM field

$$S_{free} = -mc^2 \int dt \sqrt{1-v^2/c^2}$$

EM field must be described by some 4-tensor: 4-scalar is not present in nature but 4-vector exist: 4-potential

$$A^\alpha = (\varphi/c, \vec{A}) \quad , \quad A_\alpha = (\varphi/c, -\vec{A})$$

φ = scalar potential, $\vec{A}$ = vector potential.

Superposition principle: particle in a field $A_1^\alpha + A_2^\alpha$ experiences force = sum of forces due to A_1^α and A_2^α separately.

Now we need to form 4-scalar out of particle and EM field

$$A_\alpha dx^\alpha = A_\alpha u^\alpha d\tau$$

$$\Rightarrow S_{interaction} = -q \int A_\alpha dx^\alpha = -q \int A_\alpha u^\alpha d\tau$$

q = electric charge

$$S_{int} = -q \int A_\alpha \frac{dx^\alpha}{dt} dt$$

$$\frac{dx^\alpha}{dt} = \frac{(c\,dt, d\vec{r})}{dt} = (c, \vec{v}) \Rightarrow A_\alpha \frac{dx^\alpha}{dt} = \frac{\varphi}{c} \cdot c - \vec{A} \cdot \vec{v}$$

$$\Rightarrow S_{free} + S_{int} = \int dt \left(-mc^2\sqrt{1-v^2/c^2} - q\varphi + q\vec{v} \cdot \vec{A} \right) \equiv \int dt\, L$$

looks like P.E. in potential φ

new term, depends on velocity of the particle!

$$\text{Momentum } \vec{P} = \frac{\partial L}{\partial \dot{\vec{r}}} = \frac{\partial L}{\partial \vec{v}} = \frac{m\vec{v}}{\sqrt{1-v^2/c^2}} + q\vec{A} = \gamma m \vec{v} + q\vec{A}$$

Canonical momentum

kinematic momentum $\vec{p}$

new!

Canonical momentum can also be obtained by differentiating action along the optimal trajectory over its end-point

$$-\frac{\partial}{\partial x^\alpha}\left(S_{free}+S_{int}\right) = p_\alpha + q A_\alpha = P_\alpha = (P_0, -\vec{P})$$

$$\downarrow$$

$$m u_\alpha = m\gamma(c;\vec{v})$$

$$q A_\alpha = q(\varphi/c, -\vec{A})$$

$$\Rightarrow P_\alpha = \left(mc\gamma + q\varphi/c\,,\; -m\gamma\vec{v} - q\vec{A}\right)$$

$$\Rightarrow c P_0 = \gamma mc^2 + q\varphi = \mathcal{E} \text{ energy}$$

Check: if L does not depend on t (so that $\varphi = \varphi(\vec{r})$, $\vec{A} = \vec{A}(\vec{r})$ only)

$$\Rightarrow \mathcal{E} = \vec{v}\,\frac{\partial L}{\partial \vec{v}} - L = \vec{v}\cdot\left(\gamma m\vec{v} + q\vec{A}\right) - \left(-\frac{mc^2}{\gamma} - q\varphi + q\vec{v}\cdot\vec{A}\right)$$

$$= \underbrace{\gamma m v^2 + \frac{mc^2}{\gamma}} + q\varphi = \gamma mc^2 + q\varphi \text{ indeed.}$$

$$\gamma\left(mv^2 + \frac{mc^2}{\gamma^2}\right) = \gamma mc^2$$

Equation of motion [for the particle, field A^α is held fixed]

$$\frac{d}{dt}\frac{\partial L}{\partial \vec{v}} = \frac{\partial L}{\partial \vec{r}} \Rightarrow \frac{d}{dt}\vec{P} = \frac{\partial L}{\partial \vec{r}}$$

canonical momentum

$$\vec{P} = \vec{p} + q\vec{A}$$

$$\frac{\partial L}{\partial \vec{r}} = \frac{\partial}{\partial \vec{r}}\left(-q\vec{\varphi} + q\vec{v}\cdot\vec{A}\right) = -q\vec{\nabla}\varphi + q\vec{\nabla}(\vec{v}\cdot\vec{A})$$

Note that velocity of particle $\vec{v}(t)$ is not function of $\vec{r}$

$\Rightarrow$ in $\vec{\nabla}(\vec{v}\cdot\vec{A})$ = derivative acts on $\vec{A}$ only.

Once derivatives are taken, $\varphi(\vec{r},t)$ and $\vec{A}(\vec{r},t)$ need to be taken at $\vec{r} = \vec{r}(t)$ where particle is at time t.

$$\Rightarrow \frac{d}{dt}\left(\vec{p} + q\vec{A}\right) = -q\vec{\nabla}\varphi + q\,\vec{\nabla}(\vec{v}\cdot\vec{A})$$

$\Rightarrow$ for the motion of the particle

$$\frac{d\vec{p}}{dt} = -q\frac{d\vec{A}}{dt} - q\vec{\nabla}\varphi + q\,\vec{\nabla}(\vec{v}\cdot\vec{A})$$

But $$\frac{d\vec{A}(t,\vec{r})}{dt} = \frac{\partial\vec{A}}{\partial t} + \frac{\partial\vec{A}}{\partial\vec{r}}\cdot\frac{\partial\vec{r}}{\partial t} = \frac{\partial\vec{A}}{\partial t} + \vec{v}\cdot\vec{\nabla}\vec{A}$$

$$\frac{d\vec{p}}{dt} = -q\vec{\nabla}\varphi - q\frac{\partial\vec{A}}{\partial t} + q\,\vec{\nabla}(\vec{v}\cdot\vec{A}) - q\,\vec{v}\cdot\vec{\nabla}\vec{A}$$

Note: $\vec{v}\cdot\vec{\nabla}\vec{A} = \frac{\partial x^j}{\partial t}\frac{\partial A^i}{\partial x^j}$ (for $i = 1,2,3$)

$$\vec{\nabla}(\vec{v}\cdot\vec{A}) - \vec{v}\cdot\vec{\nabla}\vec{A} = \vec{e}_i\frac{\partial}{\partial x_i}v_j A_j - v_j\frac{\partial}{\partial x_j}A_i\vec{e}_i$$

here i,j are Euclidian indices $(1,2,3)$

Check $$\vec{v}\times\underbrace{(\vec{\nabla}\times\vec{A})}_{\varepsilon_{ijk}\frac{\partial}{\partial x_j}A_k} = \varepsilon_{\alpha\beta i}v_\beta\,\varepsilon_{ijk}\frac{\partial A_k}{\partial x_j}\vec{e}_\alpha \Rightarrow$$

But $$\varepsilon_{\alpha\beta i}\,\varepsilon_{ijk} = \varepsilon_{i\alpha\beta}\,\varepsilon_{ijk} = \delta_{\alpha j}\delta_{\beta k} - \delta_{\alpha k}\delta_{\beta j}$$

$$\Rightarrow \vec{e}_\alpha\left(\delta_{\alpha j}\delta_{\beta k} - \delta_{\alpha k}\delta_{\beta j}\right)v_\beta\frac{\partial A_k}{\partial x_j} =$$

$$= \left(\delta_{\alpha j}v_\beta\frac{\partial A_\beta}{\partial x_\alpha} - v_\beta\frac{\partial}{\partial x_\beta}A_k\delta_{\alpha k}\right)\vec{e}_\alpha$$

$$= \vec{e}_\alpha\frac{\partial}{\partial x_\alpha}(v_\beta A_\beta) - v_\beta\frac{\partial}{\partial x_\beta}A_\alpha\vec{e}_\alpha$$

$\Rightarrow$ we have proven identity $\vec{v}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{v}\cdot\vec{A}) - (\vec{v}\cdot\vec{\nabla})\vec{A}$

$$\Rightarrow \frac{d\vec{p}}{dt} = q\left(-\frac{\partial\vec{A}}{\partial t} - \vec{\nabla}\varphi\right) + q\,\vec{v}\times(\vec{\nabla}\times\vec{A})$$

But $\vec{E} = -\frac{\partial\vec{A}}{\partial t} - \vec{\nabla}\varphi$, $\vec{B} = \vec{\nabla}\times\vec{A}$

$\Rightarrow$ Lorentz force $\vec{F}_L = q\vec{E} + q\vec{v}\times\vec{B}$.

$$\frac{d\vec{p}}{dt} = \vec{F}_L.$$

Observe $\vec{v}\cdot d\vec{p} = \vec{v}\cdot\vec{F}_L\, dt$

But $\vec{v}\cdot d\vec{p} = \vec{v}\cdot d(m\gamma\vec{v}) = \vec{v}\cdot\left[\gamma^3 \overset{m}{\vee} \vec{v}\,\frac{(\vec{v}\cdot d\vec{v})}{c^2} + m\gamma\, d\vec{v}\right] =$

$$\left(d\gamma = d\frac{1}{\sqrt{1-v^2/c^2}} = -\frac{1}{2}\frac{-2\vec{v}\cdot d\vec{v}/c^2}{\left(\sqrt{1-v^2/c^2}\right)^3} = \gamma^3\frac{\vec{v}\cdot d\vec{v}}{c^2} \right)$$

$$= m\gamma\left[\gamma^2 v^2\frac{\vec{v}\cdot d\vec{v}}{c^2} + \vec{v}\cdot d\vec{v}\right] = m\gamma\underbrace{\left(1+\gamma^2\frac{v^2}{c^2}\right)}_{1+\frac{v^2/c^2}{1-v^2/c^2} = \frac{1}{1-v^2/c^2} = \gamma^2}\vec{v}\cdot d\vec{v} = m\gamma^3\vec{v}\cdot d\vec{v}$$

Therefore $\vec{v}\cdot d\vec{p} = mc^2\gamma^3\frac{\vec{v}\cdot d\vec{v}}{c^2} = d(mc^2\gamma)$

Hence $d(mc^2\gamma) = \vec{v}\cdot(q\vec{E} + q\vec{v}\times\vec{B})\,dt = q\vec{E}\cdot\vec{v}\,dt =$

$$= q\vec{E}\cdot d\vec{r} = -q\underline{\vec{\nabla}\varphi}\cdot d\vec{r} = -q\,d\varphi$$

$$\Rightarrow d(mc^2\gamma + q\varphi) = 0$$

Note that this only works for the constant EM field, when $\frac{\partial\vec{A}}{\partial t} = 0$ (since then $\vec{E} \equiv -\frac{\partial\vec{A}}{\partial t} - \vec{\nabla}\varphi \to -\vec{\nabla}\varphi$).

* Constant magnetic field does not do work.

$\Rightarrow$ $\mathcal{E} = mc^2\gamma + q\varphi$ energy is conserved.

But is $\frac{d\vec{p}}{dt} = \vec{F}_L$ Lorentz invariant?

$\Rightarrow$ introduce covariant 4-dimensional formalism

$$S = S_{free} + S_{int} = -mc^2 \int d\tau - q \int A_\alpha dx^\alpha$$

Variation $x^\alpha(\tau) \to x^\alpha(\tau) + \delta x^\alpha(\tau)$

$$\delta(d\tau) = \frac{1}{c}\,\delta\left(\sqrt{dx_\alpha dx^\alpha}\right) = \frac{1}{c}\,\frac{dx_\alpha\,\delta(dx^\alpha)}{\sqrt{dx_\alpha dx^\alpha}} = \frac{1}{c}\,\frac{dx_\alpha\,\delta(dx^\alpha)}{c\,d\tau} =$$

$$= \frac{1}{c^2}\,u_\alpha\,\delta(dx^\alpha) = \frac{1}{c^2}\,u_\alpha\,d(\delta x^\alpha)$$

Note that $\delta(dx^\alpha) = \delta\left(x^\alpha(\tau+d\tau) - x^\alpha(\tau)\right) =$

$$= \delta x^\alpha(\tau+d\tau) - \delta x^\alpha(\tau) = d(\delta x^\alpha)$$

$$\Rightarrow \delta S_{free} = -mc^2 \int \frac{u_\alpha}{c^2}\,d(\delta x^\alpha) = -m\int\Big\{d(u_\alpha\,\delta x^\alpha) -$$

$$- du_\alpha\,\delta x^\alpha\Big\} = m\int du_\alpha\,\delta x^\alpha = m\int d\tau\,\frac{du_\alpha}{d\tau}\,\delta x^\alpha.$$

(but $\int_1^2 d(u_\alpha\,\delta x^\alpha) = u_\alpha\,\delta x^\alpha\Big|_1^2 = 0$ since $\delta x^\alpha\Big|_1^2 = 0$.)

Next, $\delta S_{int} = -q\int \delta A_\alpha\,dx^\alpha + A_\alpha\,\delta(dx^\alpha) =$

$$= -q\int\left(\frac{\partial A_\alpha}{\partial x^\beta}\,\delta x^\beta\,dx^\alpha + \underbrace{A_\alpha\,d(\delta x^\alpha)}_{\underbrace{d(A_\alpha\,\delta x^\alpha)}_{\text{full derivative again}} - dA_\alpha\,\delta x^\alpha}\right) =$$

$$= -q\int \frac{\partial A_\alpha}{\partial x^\beta}\,\delta x^\beta\,dx^\alpha + q\int \frac{\partial A_\alpha}{\partial x^\beta}\,dx^\beta\,\delta x^\alpha$$

$$= q\int \frac{\partial A_\alpha}{\partial x^\beta}\left(dx^\beta\,\delta x^\alpha - \delta x^\beta\,dx^\alpha\right)$$

(2.2)

But $\frac{\partial A_\alpha}{\partial x^\beta}\,\delta x^\beta\, dx^\alpha = \frac{\partial A_\beta}{\partial x^\alpha}\,\delta x^\alpha\, dx^\beta$ (just re-naming indices)

$$\Rightarrow \delta S_{int} = q\int \left(\frac{\partial A_\alpha}{\partial x^\beta} - \frac{\partial A_\beta}{\partial x^\alpha}\right) dx^\beta\, \delta x^\alpha =$$

$$= q\int d\tau \left(\frac{\partial A_\alpha}{\partial x^\beta} - \frac{\partial A_\beta}{\partial x^\alpha}\right) u^\beta\, \delta x^\alpha$$

$$\Rightarrow \delta S = \int d\tau \left(m\frac{du_\alpha}{d\tau} + q\, F_{\beta\alpha}\, u^\beta\right)\delta x^\alpha$$

$$\Rightarrow \quad m\frac{du_\alpha}{d\tau} = q\, F_{\alpha\beta}\, u^\beta \qquad (m \times \text{4-acceleration} = \text{4-force})$$

$$F_{\alpha\beta} = \frac{\partial A_\beta}{\partial x^\alpha} - \frac{\partial A_\alpha}{\partial x^\beta}$$ electromagnetic <u>field tensor</u>

Field tensor is antisymmetric $F_{\alpha\beta} = -F_{\beta\alpha}$

$$\Rightarrow F_{\alpha\beta} = \begin{pmatrix} 0 & \times & \times & \times \\ \times & 0 & \times & \times \\ \times & \times & 0 & \times \\ \times & \times & \times & 0 \end{pmatrix}$$

$\rightarrow$ 6 off-diagonal elements

$\|$

components of 3-dim. vectors $\vec{E}$ and $\vec{B}$.

$F_{00} = 0 \Rightarrow$ Consider $F_{0i} = \frac{\partial A_i}{\partial x^0} - \frac{\partial A_0}{\partial x^i} = \frac{1}{c}\frac{\partial}{\partial t}(-A_i) - \frac{1}{c}\frac{\partial}{\partial x^i}\varphi$

$i = 1,2,3$

$$A_\alpha = (\varphi/c, -\vec{A}) \qquad = \frac{1}{c}\left(-\frac{\partial A_i}{\partial t} - \frac{\partial \varphi}{\partial x^i}\right)$$

$$= \frac{1}{c}E_i$$ i-th component of el. field.

$$F_{12} = \frac{\partial A_2}{\partial x^1} - \frac{\partial A_1}{\partial x^2} = -\frac{\partial A_y}{\partial x} + \frac{\partial A_x}{\partial y} = -(\vec{\nabla}\times\vec{A})_z = -\vec{B}_z$$

but $(\vec{\nabla}\times\vec{A})_3 = \frac{\partial}{\partial x}A_y - \frac{\partial}{\partial y}A_x$

Similarly $F_{13} = B_y$, $F_{23} = -B_x$

$$\Rightarrow F_{\alpha\beta} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

covariant field tensor

$$F^{\alpha\beta} = g^{\alpha\gamma} g^{\beta\delta} F_{\gamma\delta}$$

$$g^{\alpha\gamma} = g_{\alpha\gamma} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$\Rightarrow$ raising (or lowering) spatial index ($i = 1,2,3$) changes the sign of the component.
Time component does not change.

$\Rightarrow$ obtain $F^{\alpha\beta}$ by changing signs of all space-time components (F_{0i} and F_{i0}) while keeping space components (F_{ij} with $i,j \in 1,2,3$) the same.

$$\Rightarrow F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

[Brau's book: Chap. 1.4 + Chap. 2.2]

Back to $m \frac{du_\alpha}{d\tau} = q F_{\alpha\beta} u^\beta$

Use: $d\tau = dt/\gamma$, $u^\beta = \gamma(c, \vec{v}) = \gamma(c, v_x, v_y, v_z)$, $u_\alpha = \gamma(c, -\vec{v})$

$\underline{\alpha=1}$ $\Rightarrow$ $m\gamma \frac{d}{dt}(-\gamma v_x) = q\gamma F_{1\beta} u^\beta =$

$$= q\gamma\left(F_{10} u^0 + F_{12} u^2 + F_{13} u^3\right)$$

$$= q\gamma\left(-\frac{E_x}{c} c + (-B_z) v_y + B_y v_z\right)$$

$$= q\gamma\left(-E_x - B_z v_y + B_y v_z\right)$$

$$\Rightarrow \frac{d}{dt}(m\gamma v_x) = q\left(E_x + v_y B_z - v_z B_y\right)$$

$$\Rightarrow \frac{d}{dt}(\vec{p})_x = q\left(\vec{E} + \vec{v} \times \vec{B}\right)_x$$ $\underline{\text{ok}}$

$\underline{\alpha=0}$ $m\gamma \frac{d}{dt}(\gamma c) = q\gamma F_{0\beta} u^\beta = q\gamma\left(F_{01} u^1 + F_{02} u^2 + F_{03} u^3\right)$

$$= q\gamma \frac{1}{c}\left(E_x v_x + E_y v_y + E_z v_z\right)$$

$$\Rightarrow m \frac{d}{dt}(\gamma c^2) = q \vec{E} \cdot \vec{v}$$

$$\underbrace{\frac{d}{dt}(mc^2\gamma)}_{\frac{d\mathcal{E}}{dt}} = q \vec{v} \cdot \vec{E}$$

$\mathcal{E} = mc^2\gamma$ = total mechanical energy